

RUSHIKESH MOHALKAR

AI/ML ENGINEER

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PROFESSIONAL SUMMARY

AI/ML Engineer focused on Generative AI, Deep Learning, Large Language Models, and production AI systems. Hands-on experience building Retrieval-Augmented Generation (RAG) pipelines, multi-agent systems, LLM applications, self-hosted AI systems, and machine learning solutions using Python, PyTorch, Hugging Face, LangChain, LangGraph, FastAPI, Docker, and AWS. Strong software engineering foundation with experience in data processing, automation, SQL, Git, CI/CD, APIs, and enterprise application development.

TECHNICAL SKILLS

Programming: Python, Java, SQL, Bash

Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Feature Engineering, Data Preprocessing, Model Evaluation

Deep Learning: PyTorch, TensorFlow, Keras, CNNs, RNNs, LSTM, Transfer Learning

Generative AI: LLMs, RAG, Prompt Engineering, LangChain, LangGraph, LlamaIndex, Hugging Face, Ollama, QLoRA, AI Agents, Multi-Agent Systems

NLP: NLTK, Named Entity Recognition (NER), Text Classification, Sequence Modeling

AI/ML Engineering: FastAPI, Flask, Vector Databases, FAISS, Qdrant, Embeddings, Reranking, Inference Pipelines, Model Optimization

Data & AI Infrastructure: PostgreSQL, SQLite, Redis, BM25, MLflow, Prometheus

DevOps & Cloud: Docker, Docker Compose, Nginx, Jenkins, Git/GitHub, Linux, AWS (EC2, S3), CI/CD

PROJECTS

Enterprise RAG & AI Operations Platform | [GitHub](#)

Technologies: Python, LangGraph, FastAPI, Qdrant, PostgreSQL, Redis, BM25, Claude API, Docker

- Built a production-oriented Retrieval-Augmented Generation (RAG) platform using hybrid dense and BM25 retrieval, reranking, HyDE, Corrective RAG (CRAG), and Self-RAG, achieving 0.76 Precision@5 on a labeled benchmark.
- Implemented Text-to-SQL with a human-approval workflow using LangGraph interrupts for controlled database interactions.
- Developed a nine-layer input/output guardrail pipeline, semantic caching, offline fallback capabilities, and REST APIs.
- Built an automated evaluation framework with 160 tests to validate retrieval, generation, guardrails, and system behavior.

Custodian - Multi-Agent Accounts Payable AI | [GitHub](#)

Technologies: Python, LiteLLM, FastAPI, React, MLflow, Prometheus, SQLite, Docker

- Built a governed multi-agent AI system for invoice ingestion, fraud and risk scoring, approval routing, and automated payment workflows.
- Integrated multiple LLM providers including OpenAI, Groq, Gemini, Anthropic, and Hugging Face through LiteLLM, with heuristic fallback support for offline operation.
- Implemented PII redaction, policy enforcement, append-only audit logging, OCR ingestion, role-based API authentication, MLflow experiment tracking, and Prometheus monitoring.
- Developed a React dashboard and automated testing pipeline with 95 backend tests and 33 frontend tests.

RAGApp - PDF Question Answering System | [GitHub](#)

Technologies: Python, LangChain, FAISS, Hugging Face Transformers, RapidOCR, Gradio

- Built a fully offline RAG pipeline for PDF ingestion with OCR fallback, semantic chunking, embeddings, vector retrieval, and grounded question answering.
- Implemented local model inference using Hugging Face Transformers and FAISS for efficient similarity search without external API dependencies.
- Developed an interactive Gradio interface for document-based question answering.
- Resolved offline runtime and dependency issues to maintain a stable end-to-end PDF-to-answer workflow.

Oryza - Self-Hosted LLM Chatbot | [GitHub](#)

Technologies: Python, Ollama, Qwen 2.5, Flask, Docker, Docker Compose, Nginx

- Built a private, offline chatbot serving the Qwen 2.5 1.5B language model locally through Ollama and Flask.
- Developed the application to operate without external LLM APIs, supporting privacy-focused and self-hosted inference.
- Containerized the application using Docker Compose and configured Nginx as a reverse proxy.
- Optimized the local inference workflow for deployment in resource-constrained environments.

NanoChat-X - GPT-Style Transformer Built From Scratch | [GitHub](#)

Technologies: Python, PyTorch, FastAPI

- Built a decoder-only GPT-style Transformer from scratch using causal self-attention, positional embeddings, token embeddings, and a weight-tied language model head.
- Implemented a custom tokenizer, training loop, AdamW optimization, warmup-cosine learning rate scheduling, gradient accumulation, and automatic mixed precision (AMP).
- Implemented temperature and top-k sampling for controllable text generation.
- Built a FastAPI interface for interactive inference and validated causal masking through automated regression tests.

PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions | Programmer Analyst | Jul 2025 - Present

- Developed Python-based data processing and automation solutions, streamlining engineering workflows and reducing manual effort by 15%.
- Built and maintained data-driven workflows and reporting solutions, improving process efficiency and reducing delivery delays by 20%.
- Worked with Python, SQL, Git, Docker, Jenkins, and CI/CD workflows in an enterprise engineering environment.

Cognizant Technology Solutions | Programmer Analyst Trainee (PAT) | Jul 2024 - Jul 2025

- Developed software automation and data-processing solutions using Java, Python, SQL, Git, Docker, and Jenkins.
- Implemented reusable Python and Java components to automate workflows and improve engineering efficiency.
- Collaborated with engineering teams to develop, troubleshoot, and maintain scalable software solutions across enterprise applications.

Cognizant Technology Solutions | Intern | Jan 2024 - Apr 2024

- Developed Java and Python-based software components and automation workflows using SQL, Git, Maven, and Docker.
- Built reusable scripts and application workflows using object-oriented programming, data processing, and version control.
- Worked in an Agile engineering environment, contributing to application development and continuous improvement.

EDUCATION

B.E. - Electronics & Telecommunications Engineering | AISSMS Institute of Information Technology, Pune | 2019 - 2023

CGPA: 8.21 / 10

CERTIFICATIONS & LEARNING

- [Mathematical Foundations of Machine Learning](#)
- [Python for Data Science & ML Bootcamp](#)
- [Deep Learning A-Z - Neural Networks, AI & ChatGPT](#)
- [AI Engineer Core Track - LLMs, RAG, QLoRA & Agents \(In Progress\)](#)
- [Generative AI with LangChain & Hugging Face \(In Progress\)](#)